Marco **Valle**

Back-End Developer & Data Analyst

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Summary

Computer Science graduate (BSc, 99/110) specialized in back-end development (Python, Django, SQL) and data analysis (PyTorch, NumPy, pandas). Built a Django ERP now in production at Aquaforma S.r.l. and a Transformer model predicting aircraft trajectories from 65M ADS-B records.

Skills

Programming	Python, SQL, C/C++, Kotlin, JavaScript (basic)
Back-End & Web	Django (ORM, migrations), PostgreSQL, SQLite, HTML, CSS
Data Analysis & ML	PyTorch, NumPy, pandas, Matplotlib; data pipelines, model training & evaluation
DevOps & Cloud	Git, GitHub Actions (CI), Docker & Docker Compose, Gunicorn, Caddy, Linux VPS deployment
Embedded & Mobile	ESP32/Arduino (C/C++), I2C/ADC sensors, Bluetooth; Android (Kotlin), Gradle
Languages	Italian (native), English (B2, certified)

Education

University of Milano-Bicocca — *Milan, Italy*

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY (L-31) — *Sep. 2021 – Mar. 2025*

- Graduated with **99/110**. Thesis: *Non-Autoregressive Models for Air Traffic Trajectory Prediction* (PyTorch).

University of Łódź — *Łódź, Poland*

ERASMUS+ EXCHANGE PROGRAMME — *Oct. 2023 – Mar. 2024*

Experience

Aquaforma S.r.l. — *Treviolo, Italy*

PARTNER AND EMPLOYEE — *2026 – Present*

- Administer and customize **Odoo Enterprise (ERP/CRM)**; manage IT infrastructure, network and software systems; maintain the corporate website.
- Analyse business needs and develop the in-house **ERP in Python/Django** (see Projects), now the core management system.

Pavis S.r.l. — *Rovato, Italy*

SWIMMING POOL INSTALLATION AND MAINTENANCE — *Mar. 2023 – 2025*

- Seasonal pool maintenance including water-chemistry analysis (pH, chlorine, cyanuric acid, salinity); customer relations and on-site coordination.

Poppies S.r.l. — *Milan, Italy*

IT SUPPORT AND CONSULTING — *Feb. 2023 – Jun. 2023*

- Configured on-premise network services and local data infrastructure; maintained and upgraded workstations and hardware.

Projects

In production at Aquaforma S.r.l.

BUSINESS MANAGEMENT SYSTEM (ERP) — *2026 – Present*

- Designed and built a full-stack ERP in **Python/Django + PostgreSQL**: contacts, catalogue, quotes, orders, invoices, inventory, supplier replenishment, **Italian e-invoicing (SDI)** with FatturaPA XML (FPR12) and PEC transmission.
- Deployed on Linux VPS with **Docker Compose** (Gunicorn, Caddy, HTTPS, backups); **480+ automated tests**, **CI on GitHub Actions**; read-only **MCP server** for AI-assisted data access.

Personal Project

FLIGHTPREV — AIRCRAFT TRAJECTORY PREDICTION — *2025 – 2026*

- Built a **PyTorch Transformer** predicting aircraft positions 10–120 s ahead from **65M ADS-B records** (full data pipeline: cleaning, segmentation, resampling, sliding windows).
- Reached **582 m average ADE vs 833 m** of the best kinematic baseline (**+30%**) via residual learning; outperformed a **FlightBERT** reimplementation by 75–90% at all horizons.

Personal Project

IOT HOME AUTOMATION & POOL WATER MONITORING — *Ongoing*

- ESP32 firmware (C/C++/Arduino) for relays and pool water chemistry (I2C, 16-bit ADC); **Android (Kotlin)** app with Bluetooth control.

Extracurricular Activities

Scout Educator, Federazione Scout Europa (Bergamo) — Sep. 2019 to present: plan and lead educational activities for teens aged 15–19; leadership and teamwork.